Collection Framework

Framework Fusion: Mastering Java



Learning Outcome

1. Understand the concept of a framework
2. Explain the purpose and structure of the Java Collection Framework
3. Identify the limitations of arrays and how collections solve them.

Before Starting ,Lets Recall

- Basic Java concepts: OOPs (class, object, inheritance, polymorphism)
- Wrapper classes & Autoboxing
- Equals() and hashCode() methods
- Iteration concepts (for-each loop)

What is a Framework in Java?



A framework is a predefined structure or set of reusable classes and interfaces that helps developers build applications efficiently. It acts as a skeleton or scaffolding for software development.

Key Features:



Standard Functionalities

Offers pre-built standard functionalities so developers don't have to write common code from scratch.



Tools & Libraries

Provides essential tools, comprehensive libraries, and API sets needed for development.

What is the Collection Framework?

The Java Collection Framework (JCF) is a unified architecture for storing, retrieving, and manipulating groups of objects.



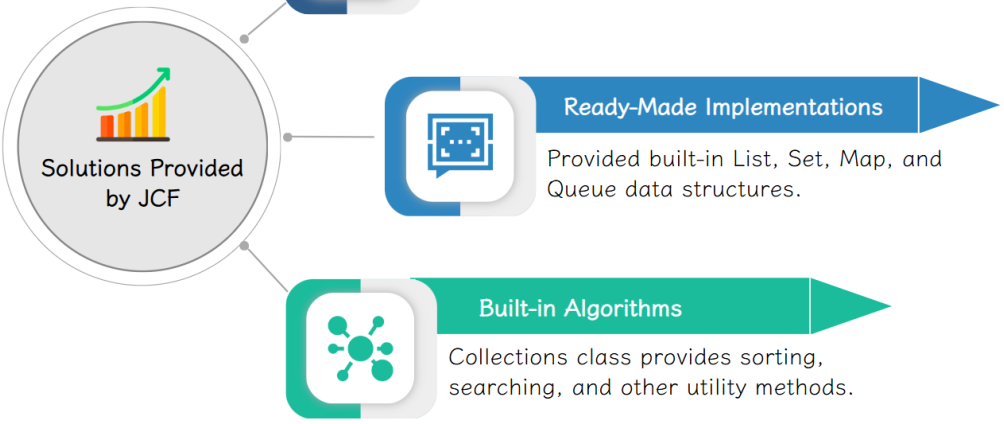
It includes:

- Interfaces (like List, Set, Queue)
- Implementing classes (like ArrayList, HashSet)
- Algorithms (like sorting and searching)

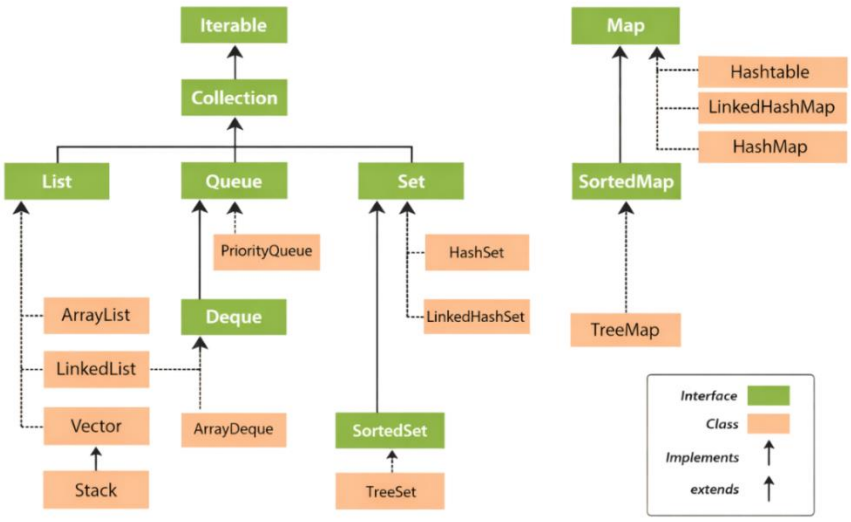


Problems Before JCF

1. **Only Arrays Were Available:**
   Java mainly used arrays, which had fixed size and were not dynamic.
2. **No Standard Data Structures:**
   There were no built-in List, Set, or Map implementations.
3. **More Coding & Complexity:**
   Developers had to manually write logic for sorting, searching, etc.



Java Collections Framework Hierarchy



Interface	Purpose
List	Ordered collection, allows duplicates
Set	No duplicate elements
Queue	Processes elements in FIFO order
Deque	Allows insertion/removal from both ends
Map	Stores key–value pairs (unique keys)
SortedSet	Set that maintains ascending order
NavigableSet	SortedSet with navigation methods
SortedMap	Map that maintains sorted keys
NavigableMap	SortedMap with navigation methods

Summary

1. Framework are reusable classes and interfaces for efficient coding.
2. JCF manages groups of objects with List, Set, Queue, Map.
3. Solves array limitations
4. Advantages: ready-made data structures and efficient algorithms.

Quiz

?

Which interface in the Collection Framework is the root of all collection types?

A. List

B. Map

C. Collection

D. Queue

Quiz-Answer

✓

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